An Exploratory Analysis of Audience Size Trends of Human and AI Art on Instagram

AP Research

Word Count: 5,496

INTRODUCTION

In recent years, the concept of art created with Artificial Intelligence (AI) has gone from being thought of as a futuristic fantasy, to a controversial but prominent topic in the field of art today. Now, anyone with access to the internet can create this art by entering text prompts into AI image generation programs/softwares such as DALLE 3, Midjourney, and Stable Diffusion. This has led to the rapid proliferation of AI-generated art, especially on the internet and social media platforms. As of 2019, according to Kang et al., 94% of online artists use Instagram for art-related purposes, which brings my focus to Instagram (Kang et al., 2019). Currently, the "#aiart" hashtag on Instagram has 14.4 million posts alone. Due to its increase, many in the art world worry that AI art profiles may take over the position that traditional human artists have held online. If this is true, the rise of AI art would have left human artists with a smaller audience. Current research has shown that audiences drive engagement, interaction, and creators' voices. Without their audiences, human artists may lose their ability to be compensated for their art and a platform to communicate their ideas to an audience (Li & Kang, 2021; Kang et al., 2019). If AI art negatively affects human artists, it's important to identify this problem. Unfortunately, the topic of AI art's real-world effect or even individual performance trends on social media has received very limited attention. Data surrounding this topic can be complex and tends to be qualitative, such as user audiences, artist profiles, art genres, concurrent real-world trends, and aesthetic appeal of posts. Proving causation in this relationship is challenging, even more so since this topic is very early in its development and lacks coverage in current literature. With this in mind, this paper aims to provide an exploratory analysis of historical human and AI art trends of audience size on Instagram. I hypothesize that the audience size trend of human art

is negatively associated with the audience size trend of AI art over time, with a decrease in the growth rate of human art and increase in that of AI art in recent years.

Definition of Key Terms:

AI Art: 2D art images created by softwares that use artificial intelligence to generate images from text prompts, such as DALLE 3, Midjourney, and Stable Diffusion.

AI Artist: A human who explicitly uses generative AI software to create AI art images.

Human Art: 2D art images created by humans without the use of generative artificial intelligence. This includes traditional mediums such as paint or pencil, digital mediums, such as softwares like Clip Studio Paint, Krita, and Photoshop (not including Generative AI functionality), or a combination of both.

LITERATURE REVIEW

One of the driving reasons behind my hypothesis is research showing human bias to AI art (Chamberlain et al., 2018; Millet et al., 2023; Bellaiche et al., 2023). These articles focus on human reactions against AI art, whether humans are biased to like or dislike art if it is supposedly labeled as generated by AI. Chamberlain et al. studied the ability of observers to distinguish between the two types of art, but this paper was published in 2018, which may be outdated considering the staggering rate at which AI art aesthetic and image quality has progressed since then. The previously mentioned studies come to the unanimous conclusion that humans have a strong bias against AI art. This bias was especially stronger in humans with stronger anthropocentric[1] creativity beliefs. This shows that it's possible that humans with greater anthropocentric creativity beliefs feel threatened by AI art. However, Bellaiche et al. believe that this bias was more modest while changing the AI label of the art. These studies did not

[1] The idea that humans are the central/most important part of the world, in this case opposed to AI.

investigate this bias in an online context, such as on social media, which leaves more to be discovered.

There has been significant research surrounding the increasing effect of AI art on humanity. First, the development of AI art itself has been well documented, with reviews on its history, creative methods, and future outlook (Cetinic & She, 2021). Dejan Grba believes that AI raises issues regarding ethics and creativity, especially in a society that is moving towards technocentrism[2] (Grba, 2022). Joanna Zylinska also mentions the conflicting and "ethically dubious" relationship between human artists, big tech companies, and other AI artists (Zylinska, 2020). There are also investigations on the topic of AI art as a forgery of historical art from Tsila Hassine and Ziv Neeman, which contribute to the discussion of AI art on humanity as it is believed to devalue other art (Hassine & Neeman, 2019). These studies are mostly in agreement that AI art has been increasing, with beginnings in the late 1900s, rapidly developing in the later half of the 2010s, and that it has questionable ethics and implications across multiple stakeholders, often with human artists as the injured party.

These studies, along with research that targets Instagram specifically (Li & Kang, 2021; Kang et al., 2019; Jaakonmäki et al., 2017), contributed to the framework of many of the underlying assumptions present in this paper. All of these studies suggest that Instagram accounts rely on their audience to support them, which is a large part of the significance of this study. Lev Manovich situated the Instagram platform within a global context and examined how visual art's history contributes to design culture and trends alongside other forms of art (Manovich, 2017). This presents the interconnected nature of Instagram, which leads me to investigate the relationship between 2 groups of artists on Instagram.

[2] The view that technology is most valuable for it's capability to solve problems and driving change in society.

There have been many papers that have utilized machine learning methods to analyze posts on Instagram (Giannoulakis & Tsapatsoulis, 2015; Philp et al., 2022; Rivera, 2023; Yu & Egger, 2021; Arafieva et al., 2021). In their paper, Yu and Egger employed machine learning to analyze visual content, such as color, to predict post engagement. Similarly, Philp et al. also used machine learning methods to predict engagement from marketing Instagram posts, using factors such as image quality, brand centrality, and presence of people, faces, or brand logos to influence engagement. While these methodologies are too advanced to implement by myself, they provide insight into how to collect data from Instagram to be used in machine learning models and functionally conduct analysis, especially through Python.

Past research has analyzed human bias against AI art, performance and audiences of Instagram art posts, and Instagram as a social media platform for artists, but has not investigated the trends or growth of AI and human art audience sizes on Instagram in recent years, a clear gap in the research. Building off of methodologies from the aforementioned studies, I created my own study design to answer the research question: how have the audience sizes for human artists situated with AI art changed over time on Instagram? This study will address the large gap of historical trends and association in AI and human art online, considering post performance and audience sizes.

METHODOLOGY

Sampling:

My data collection started by creating a list of distinct art genres that seemed to be most prevalent in art on Instagram. These categories were not comprehensive, but rather a means of diversifying the sample. Curation came from compiling known genres in other studies, such as from Kang et al. (2019) and my personal experience on Instagram. My initial list, L_1, included the following genres: Anime, Character Design, Environment, Fantasy, Fashion, Horror, Mecha, Portrait, Semi-Realism, Stylized, Traditional Painting, Cartoon, Comic, Digital Painting, Pencil, and Pixel. This list was used to categorize human art profiles, but a different list of genres, L_2, was used for AI artists. L_2 consists of the genres from L_1, with the exclusion of Cartoon, Comic, Digital Painting, Pencil, and Pixel, and with the addition of Realistic Anime. These differences are meant to reflect the prevalence, or dearth, of the mentioned art genres for the different groups. Then additional distinctions were created for different follower sizes, which included the intervals: (1,000, 10,000], (10,000, 50,000], (50,000, 100,000], (100,000, 500,000], and over 500,000 followers. 2 spreadsheets were created for both types of artists[3]. The goal was to have 2 profiles for every combination of follower size and art genre. For example, a single art genre category should have 18 profiles, consisting of 10 human artists and 8 AI artists. There would be 2 human artist profiles and 2 AI artist profiles that have a follower count between 50,000 and 100,000 followers within it. However, the Stylized group for human artists had 3 profiles for each follower size to account for the more diverse individual art styles that could be found in this genre.

[3] These spreadsheets can be found in Appendices A & B for a complete list of profiles in my sample.

To be recorded, the profiles first had to meet a few conditions. I searched to meet these conditions on the candidate profile's main information and feed. First, every profile must have had at least 10 posts with visible likes to yield sufficient data. For the AI artists group, I would check if the profile was explicit in their use of generative AI to create the majority of their art. This can be through the profile's bio, post captions, or hashtags such as #midjourney (softwares involving AI image generation) or #aiartcommunity. For human art, I manually scanned through the user's posts to see if the majority of their artwork looked to be created by a human. If the profile was questionable about their creation methods or used AI in part with significant human editing, I did not record it. I did include a few profiles that had 1 or 2 posts experimenting with AI but had human created art for the majority of their content. Next, the profile should have been able to be mainly characterized by one of the art genres defined earlier. Once 2 spaces for a combination of art genre and follower size were filled, new entries would not be made for that combination.

The search process was based on methods described by Yu & Egger (2021) and Khurana & Knight (2021). These studies used hashtags to find their posts. The collection of profiles utilized 2 search functions: Instagram (Mobile) and Google search. Instagram was primarily used, with Google was used to widen the breadth of candidates in order to fill sparse categories. On the Instagram search, I used keywords and hashtags relating to the art genre and added the keyword "ai" to generate more AI art candidates. For example, when looking for human artist profiles that made art in the horror genre, a possible search would be "horror art" or "#horror #art". For Google search, I used the query: "site: instagram.com" followed by keywords or hashtags relevant to the needed groups. This limited the Google results to pages on Instagram, rather than other websites that may introduce unwanted influence on my data. For example, to

get AI artists who made anime art, I'd enter the query: "site: instagram.com #aiart #anime". Initially, it was intended for AI art profiles to fill the 500,000+ followers group, but during the search, it became apparent that there were not enough AI art profiles that surpassed this number of followers. In my sample, there were a handful of missing combinations of follower size and art genre for both human and AI art because no suitable profiles were found. Of the intended 96 AI and 165 human profiles, 92 AI and 159 human profiles were recorded. AI artist usernames were searched for first, during 2/3-6/2024, and human artist usernames were searched for during 2/10-18/2024. All individual post data was recorded into the dataset during 3/3-14/2024.

Once I had 2 CSVs (converted from the spreadsheets) containing the usernames of AI artists and human artists, the next step was to obtain more analytical data from Instagram through an automated process. To collect this data from Instagram, I created a script utilizing the Instaloader Python library, as well as the Selenium and BeautifulSoup (Richardson, 2007) packages in Python. From every profile username, I collected their follower count, number of posts, and data on all of their posts. For every post, I collected the number of likes, comments, shortcode/URL, date posted, if it was an Instagram reel, and the post's owner. This method of grabbing posts by profiles was inspired by Li & Kang (2021) and Kang et al. (2019).

Data Analysis:

The performance metric I used for each post was the number of likes plus the number of comments, which I will refer to as interactions, to represent audience size. While other works use engagement rate, defined as interactions divided by the number of followers (Stevanovic, 2020), I chose interactions because the purpose of this paper is to see a change in audience size, not specifically the engagement of the post. For example, suppose a profile with 10,000 followers averages 1000 interactions per post, measuring at an engagement rate of 0.1. A decrease may

occur and the profile may now average 750 interactions per post and drop to 7,500 followers, while keeping their engagement rate at 0.1, misrepresenting the decrease in audience.

All data analysis was done using Python, where data was converted into a time series format. While the original sample included 194,174 total posts, I filtered some posts for the main analysis. Posts must have been posted in 2014 or later, been image posts (not reels), and not have a hidden amount of likes. This factors out probably erroneous data that does not represent groups accurately and only includes posts that have sufficient information. This resulted in 62,071 AI and 105,897 human art posts. To create a regularly spaced time series, each data point was calculated by the mean number of interactions of posts posted between every 3 months belonging to their respective group of artists. This 3 month frequency was chosen to minimize variation in the time series and smooth out the trends while retaining a good amount of information about the trends. This produces multiple time series, for example, one for human artists' posts and one for AI artists' posts.

I decided to estimate the overall trends of the post interactions of artists, which will be defined as the growth rate. This is a linear rate at which the group of artists' audience size/interactions changes over time. In reality, the best growth model may not be linear, but rather exponential or polynomial, and much more complex. Despite this, first-order regression was used for simpler interpretability. Additionally, visual inspection of the time series shows that a linear relationship adequately describes the relationship for the majority of the segments in my analysis. To create sample growth rates, I used the Python package pwlf for piecewise linear regression (Jekel & Venter, 2019). This package produces a continuous segmented/piecewise regression with either a known number of segments or known breakpoint locations. For a known number of segments (n_b+1) or number of breakpoints (n_b), pwlf produces a set of breakpoints (b)

of size n_b, ordered as $b_1 < b_2 < \ldots < b_{nb}$. This set includes the beginning and end of the data, so with only 2 segments in the regression, the middle breakpoint will represent the shift in the data's trend. The default model fitting for pwlf defines the sum-of-square residuals as a function of a set of breakpoint locations (SSR(b)). This presents the optimization problem of finding the set of breakpoints with the minimum SSR. This is solved through Differential Evolution (Storn & Price, 1997), implemented in the Python package SciPy (Virtanen et al., 2020). After determining a set of breakpoints, fitting follows the same process for fitting with a set of known breakpoint locations, where pwlf produces the piecewise function:

$$\mathbf{y}(x) = \begin{cases} \beta_1 + \beta_2 (x - b_1) & b_1 \leq x \leq b_2 \\ \beta_1 + \beta_2 (x - b_1) + \beta_3 (x - b_2) & b_2 < x \leq b_3 \\ \vdots & \vdots \\ \beta_1 + \beta_2 (x - b_1) + \beta_3 (x - b_2) + \cdots + \beta_{n_b} (x - b_{n_b - 1}) & b_{n-1} < x \leq b_{n_b} \end{cases}$$

with the same number of model parameters as the number of breakpoints, by following the least-squares fit of a continuous piecewise linear function process detailed by Golovchenko (2004). Each model parameter (β_j), represents the change in slope from the previous to the next segment, except for β_1, which is the first segment's y-intercept. First, the popularity boom of AI will be determined if existent, and when it happened. This will be defined as the optimal middle breakpoint of a pwlf model with 2 segments. The set of breakpoints in the aforementioned model will be used to segment the following human artist groups. In my tables, I also presented the slope of the last segment as (α_{nb-1}).

The statistics of the models include the residual sum of squares (SSR), total sum of squares (SST), and Coefficient of Determination (R^2) for the goodness of fit of the model. Each parameter (β_j) has its own test statistic (t_j), significance (p-value), and standard error (SE(β_j)).

The test statistic, t_j, follows the Student's t-distribution with $n - n_b - 2$ degrees of freedom, with n as the number of data points/observations in the time series. It is important to note that these statistics are most accurate with a set of known breakpoints (Jekel & Venter, 2019). For models that have unknown breakpoint locations, the Davies test is often used to determine the existence of a breakpoint (Muggeo, 2008). This is tested for the AI art model to confirm the existence of the breakpoint using the implementation from the piecewise-regression Python package (Pilgrim, 2021).

To find an association between AI art and human art trends, I fitted pwlf with 2 segments with the AI art time series. This produced a significant breakpoint that I used to fit a model to the human art data. Since the aim of this study is to investigate the effect on human artists, subgroups of human artists will also be compared to the overall AI art trends. To conclude my design, the null hypothesis, H_0: $\beta_{n-1} = 0$, states that the change in human art growth after the increase of AI art is nonexistent, while the alternate hypothesis,
H_a: $\beta_{n-1} < 0$, states that there was a decrease in growth for human art after the increase of AI art. These were tested at a significance level of 0.05.

RESULTS

Firstly, I investigated the average interactions of all human artists and AI artists every 3 months. This represents the audience belonging to each group, without investigating the individual differences within the groups. The 2 time series are displayed in Figure 1 and a breakdown of the human art posts dataset is presented in Table 1. The model for AI artists produced a breakpoint between May and July 2021, which is consistent with the general popularity of AI art as a whole (Zylinska, 2020). The Davies test also proves the breakpoint significance with a p-value of 1.86305E-45 at an alpha of 0.05. Through visual inspection of

Figure 2, one can assume that this breakpoint correctly indicates the shift in trends for AI art, which allows me to use the various regression statistics of the model (Jekel & Venter, 2019). There is also no obvious pattern in the residuals plot (Figure 3) and an R^2 of 0.9517, so the linear model fits well and is appropriate for the data. The results of this model are displayed in Table 2. The second segment had a significant β_3 of 228.31 at a p-value of around 8.768E-21, which means that the growth rate for the average AI art post increased by 228 interactions per 3 months from the growth rate in the first segment, which was insignificantly different than 0 after July 2021. When I fitted the same set of breakpoints in the AI art model to the average human art interactions every 3 months, I obtained a new model, displayed in Figure 4. Based on the residuals plot for these 2 segments (Figure 5) and an R^2 of 0.9641, the linear model is a good fit and appropriate for the data. The results of this model are displayed in Table 3. The model has a significant β_2 that is 372.21 with a p-value of 7.928E-28. The most noticeable shift comes with the second segment. β_3 is significant at -518.112 with a p-value of 1.705E-13. This means that the estimated growth rate of the average interactions for human art posts decreased by 518 interactions after July 2021. This led to a negative growth rate (α_2) after the breakpoint, with the average interactions for human art posts predicted to be decreasing by 145 interactions every 3 months. From this, I rejected the null hypothesis that there was no change after the increase in AI art. This shows convincing evidence that there was a decrease in human art audience growth after July 2021, when AI art increased.

Figure 1. All Average Interactions Over Time

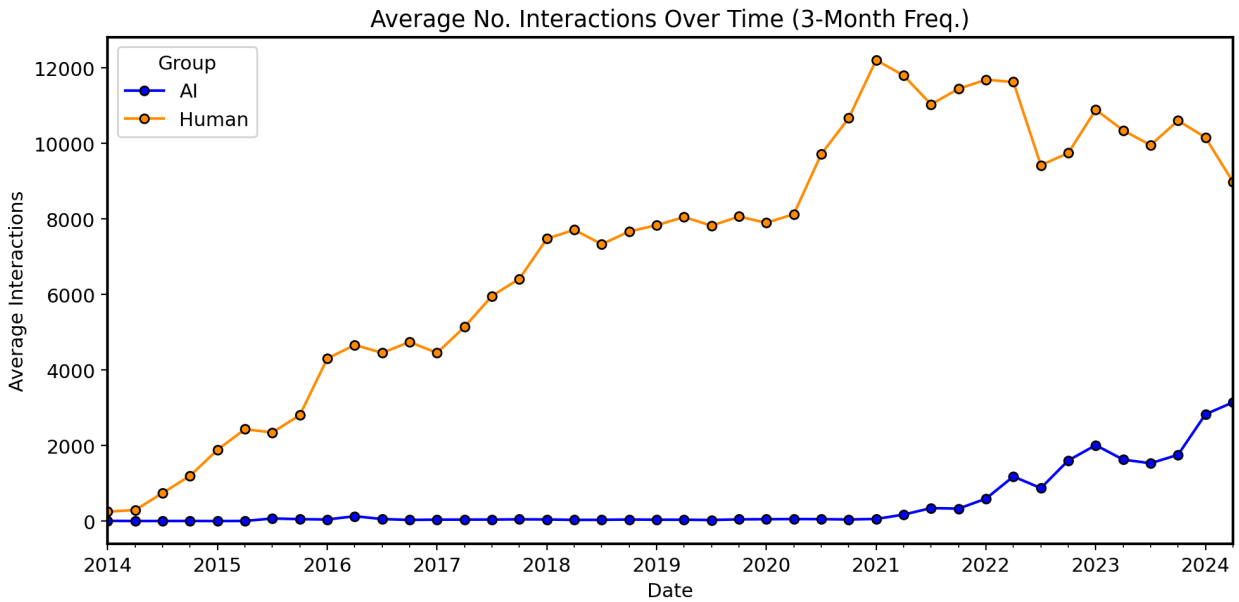


Table 1. Human Posts Dataset Groups Breakdown

Follower Size on Owner Profile	Count	Proportion of Dataset
(0, 10000]	4511	0.042597996166085915
(10000, 50000]	14876	0.14047612302520374
(50000, 100000]	14163	0.13374316552876853
(100000, 500000]	34031	0.32135943416716245
(500000, 100000000]	38316	0.3618232811127794
Art Genre	**Count**	**Proportion of Dataset**
Pixel	1448	0.013673664032031125
Digital Painting	3508	0.033126528607987
Semi Realism	3715	0.03508125820372626
Cartoon	3959	0.03738538391078123
Horror	3998	0.03775366629838428
Portrait	4034	0.03809361927155632
Anime	4566	0.04311736876398765
Mecha	5551	0.052418859835500535
Character Design	5870	0.055431220903330594
Pencil	6926	0.0654031747830439
Traditional Painting	7716	0.07286325391654154
Fantasy	8394	0.07926570157794838
Stylized	9164	0.08653691794857267
Fashion	10352	0.09775536606325014
Environment	11299	0.10669801788530364
Comic	15397	0.14539599799805472
Total	105897	1.00

Figures 2 & 3. Average AI Art Interactions Segmented Regression Model

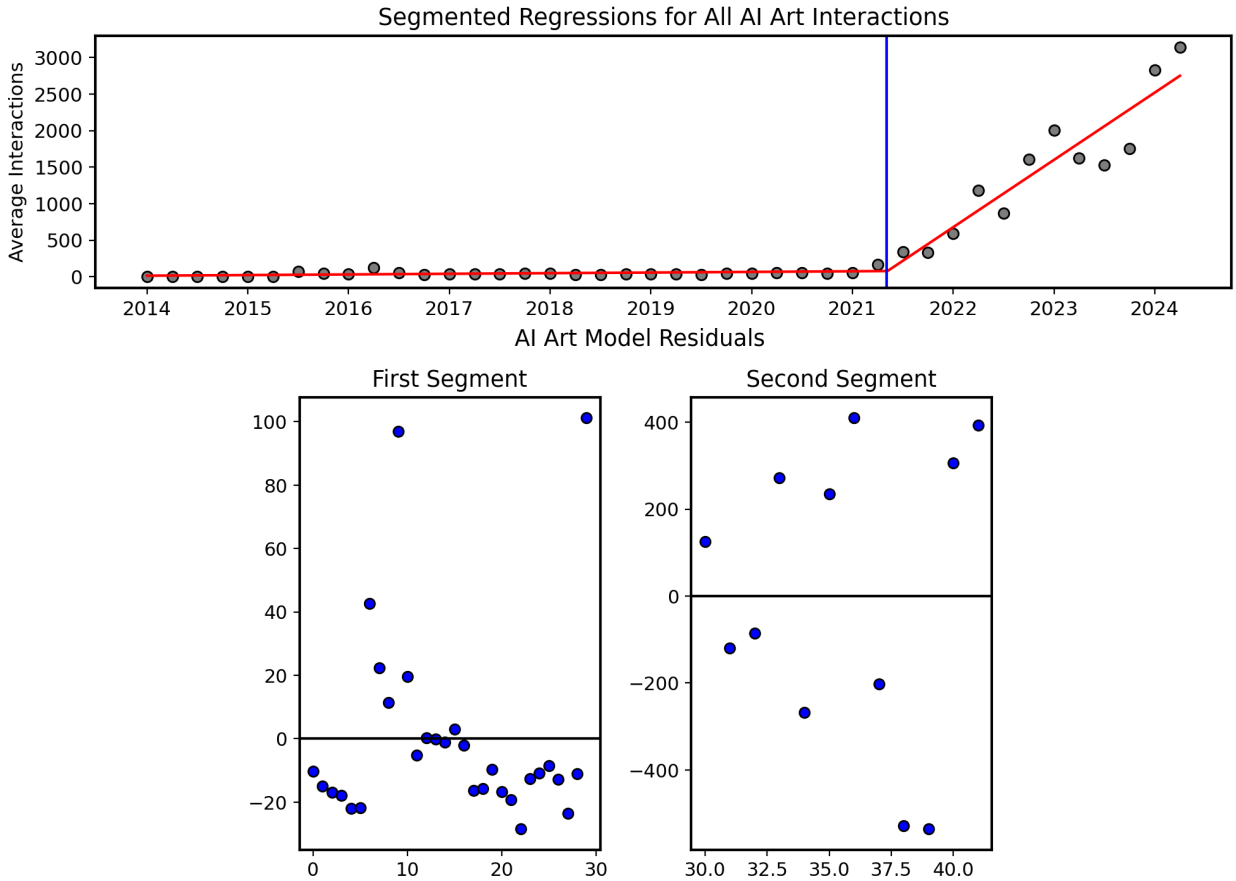


Table 2. Average AI Art Interactions Segmented Regression Results Table

No. Observations	42
No. Segments	2
SSR	1290794.486763385
SST	1356343.215198061
R^2	0.9516724618811883

Parameter	Estimate	Std. Err.	p-value
β_1	15.214228561342283	62.68413820046391	0.8095000323813689
β_2	1.888926998833412	3.421392382376811	0.5840347957827521
β_3	228.3189303755182	12.443128060468279	8.767657380596462E-21
α_2	230.2078573743516	-	-
b_2	29.35641897*	-	-

Davies test for existence of at least 1 breakpoint: p=1.86305E-45

*Ceiling of number is used as index in time series from 0 to 41, date is between 2021-04 to 2021-07

Figures 4 & 5. Average Human Art Interactions Segmented Regression Model

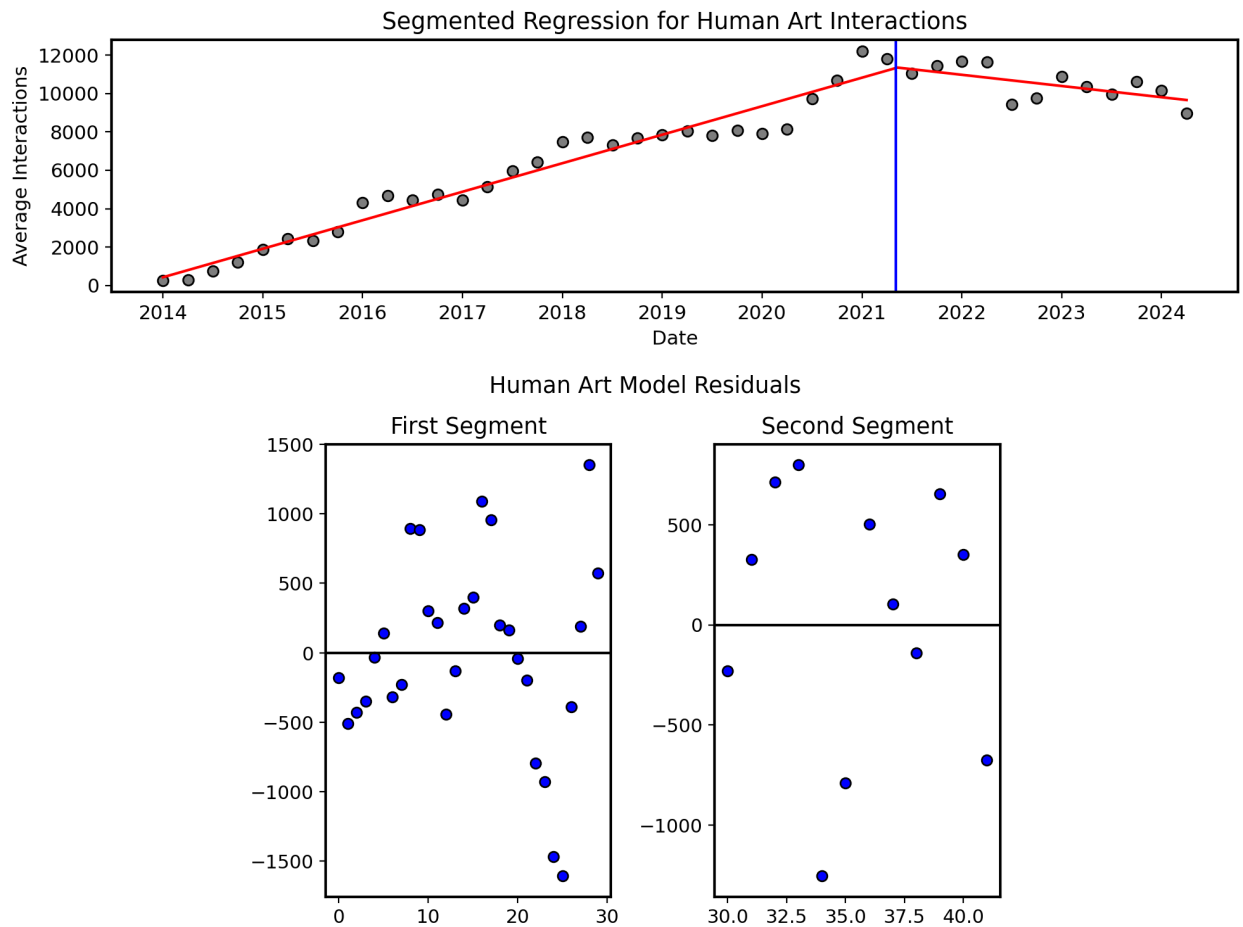


Table 3. Average Human Art Interactions Segmented Regression Results

No. Observations	42
No. Segments	2
SSR	18568070.15723647
SST	19259795.20736807
R^2	0.9640845064714411

Parameter	Estimate	Std. Err.	p-value
β_1	430.47081587618453	237.7456025276245	0.07790610702383936
β_2	372.21094741167025	12.976504372290782	7.926814226321854E-28
β_3	-518.1161838089973	47.19374092061026	1.704546957716452E-13
α_2	-145.90523639732714	-	-
b_2	29.35641897*	-	-

*Between 2021-04 to 2021-07

These results cannot be uniformly applied to all groups in the sample. In Table 1, the 2 highest follower groupings made up a very high proportion of the dataset, making up almost 70% of it. This leads to these 2 groups weighing much higher in the overall averaged model. To investigate AI's association with each group of follower size, I arranged the posts within those groups and tested each group individually. There was also an unequal proportion of art genres making up the posts, so the posts were also grouped by art genre within the groups of follower size and 3 month intervals, creating a mean that weighed each human artist combination/group of genre and follower size equally. This produced 5 different time series, one for each follower size. Based on these groups, I fitted pwlf linear models on each time series of average interactions for each group with the set of breakpoints from the initial model for the AI art, between May and July 2021. For the group with follower counts between (0, 10,000] (Figure 6, Table 4), there was a significant increase after the breakpoint, with a β_3 of 62.735 and a p-value of 1.0367E-09. For the group with follower counts between (10,000, 50,000] (Figure 5, Table 8), there was also a significant increase after the breakpoint, with a β_3 of 271.956 and a p-value of 2.5244E-18. Based on these 2 groups, I could not reject the null hypothesis. There is a clear pattern in the residuals in the (0, 10,000] model's first segment and an R^2 of 0.7004, which indicates an inadequate fit. However, through visual inspection of Figure 6, this segment would not produce a much different β_3 with a higher-order or exponential function, so I'll move past this group. While these 2 groups' increase is associated with AI art's increase, this growth is also similar to the growth of groups with higher follower sizes which was experienced in the past, as shown in Figures 13, 15, & 17. This could show that this growth is not related to AI art, but rather the natural growth that allows all profiles to reach the interactions proportional to their follower sizes.

Figures 6 & 7. Segmented Regression Model for Human Artists with (0, 10,000] Followers

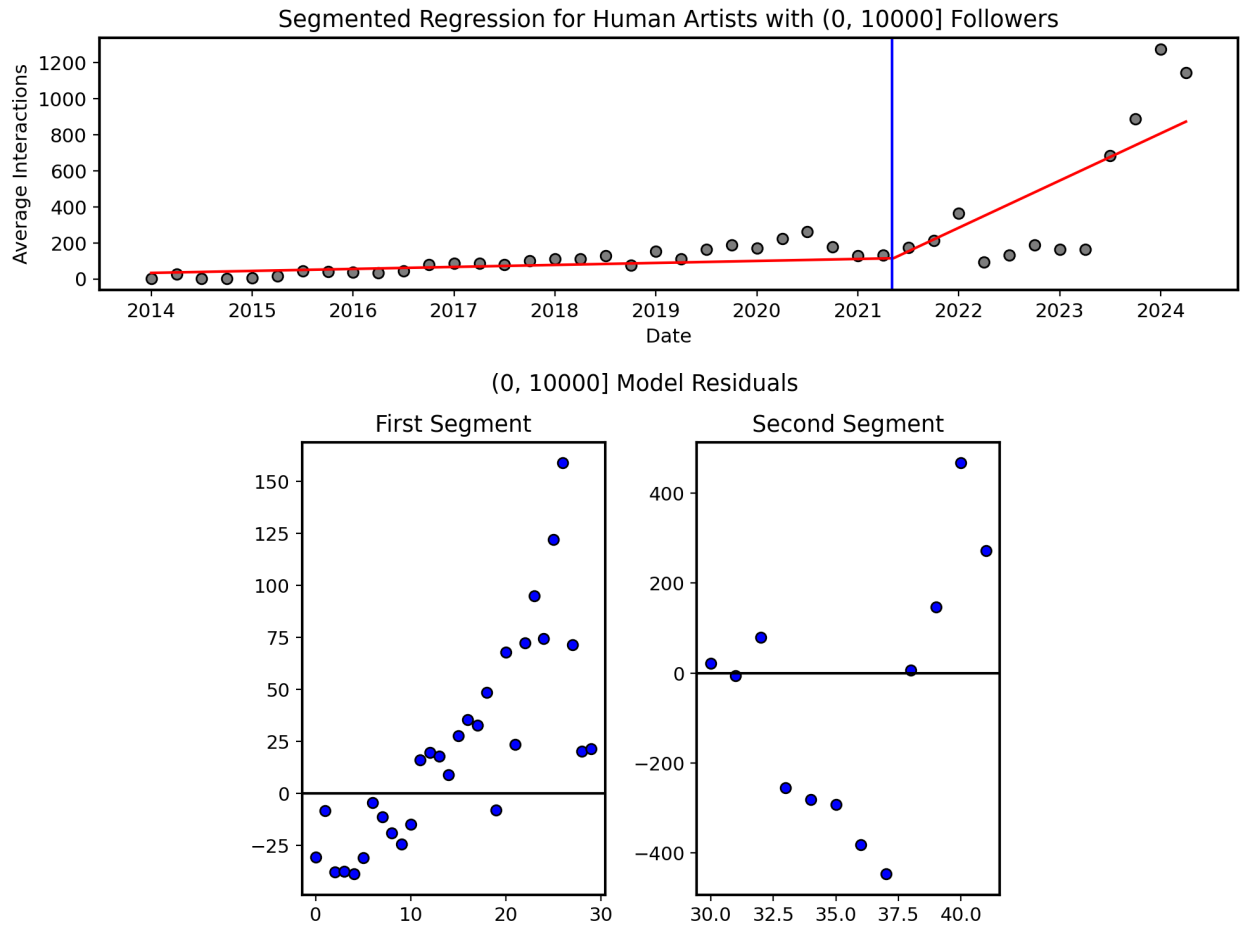


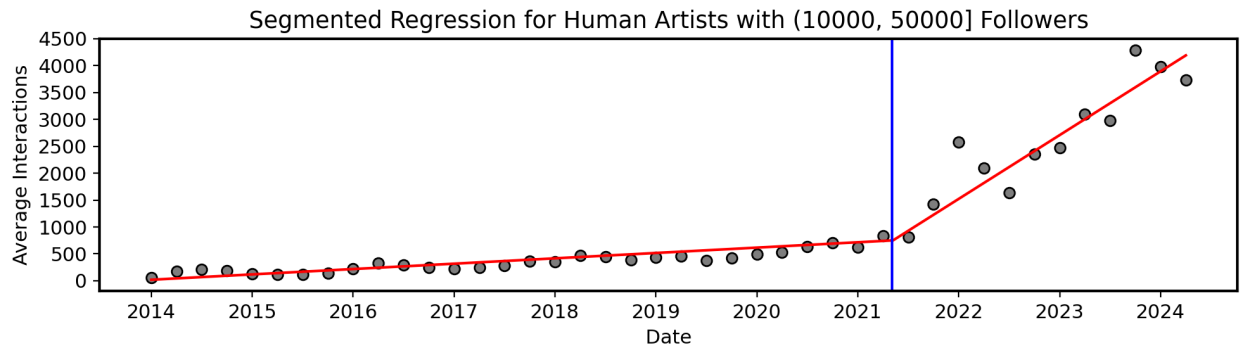
Table 4. Segmented Regression Results for Human Artists with (0, 10,000] Followers

No. Observations	42
No. Segments	2
SSR	981076.4474012506
SST	1400739.6908523424
R^2	0.7003988348500862

Parameter	Estimate	Std. Err.	p-value
β_1	34.64139839211576	54.64878782596263	0.5298522198809108
β_2	2.6443677199508695	2.9828111475797674	0.3807633735440983
β_3	62.734557211670605	10.848069177362943	1.0367006348816685E-06
α_2	65.37892493162148	-	-
b_2	29.35641897*	-	-

*Between 2021-04 to 2021-07

Figures 8 & 9. Segmented Regression Model for Human Artists with (10,000, 50,000] Followers



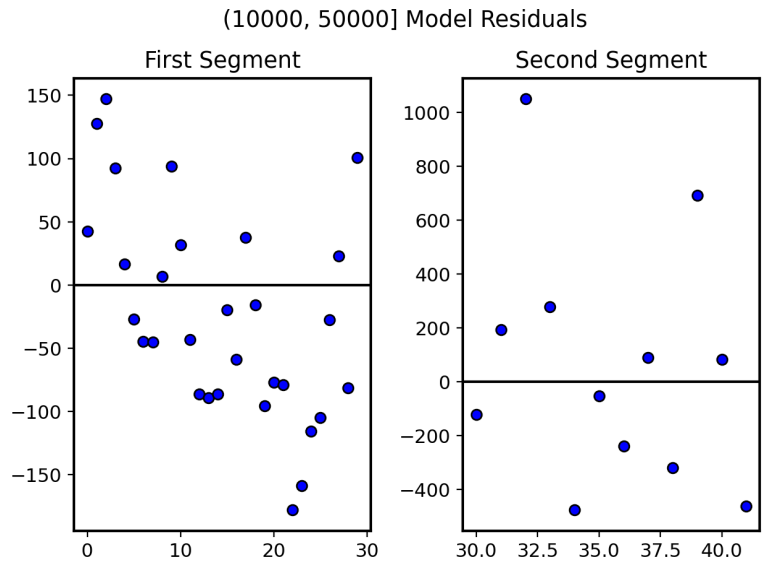


Table 5. Segmented Regression Results for Human Artists with (10,000, 50,000] Followers

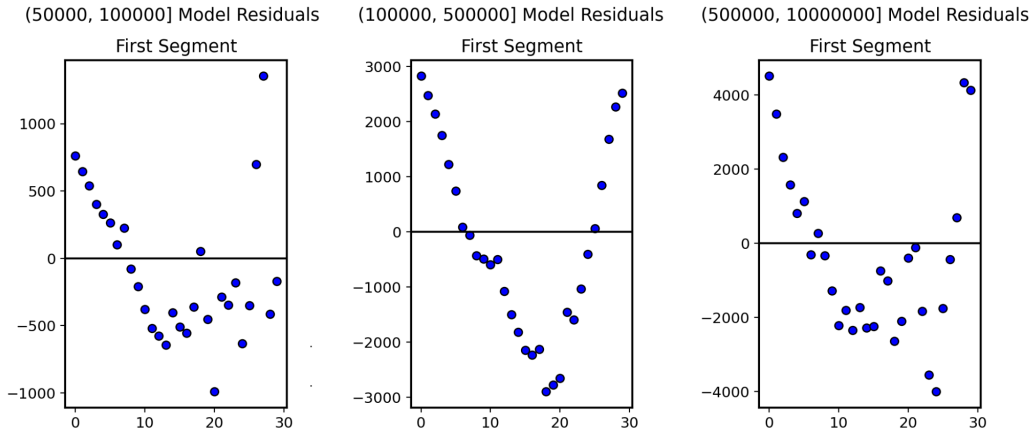
No. Observations	42
No. Segments	2
SSR	2545039.1482675504
SST	2663063.148076276
R^2	0.955681111094199

Parameter	Estimate	Std. Err.	p-value
β_1	18.768398545112518	88.01898211437158	0.8322573453434815
β_2	24.531696499947728	4.804205390346131	8.950365398613831E-06
β_3	271.9563474238358	17.47222664734294	2.5244205890275517E-18
α_2	296.4880439237835	-	-
b_2	29.35641897*	-	-

*Between 2021-04 to 2021-07

The remaining groups were fitted with 2 segments but produced first segment residuals that followed an obvious pattern (Figures 10, 11, & 12). A linear model with only 2 segments was not appropriate for the data and will misrepresent the shift at the final breakpoint. To more closely estimate the trend before the increase of AI art, an additional breakpoint was added midway between the first segment of the previous model, at the beginning of 2018. This breakpoint preserves 13 data points in the now 2nd segment, which is greater than the recommended minimum of 12 data points for segmented regression (Li et al., 2021). Fitting the remaining groups to a model with breakpoints at January 2018 and July 2021 produced my remaining results, all of which had little pattern in the residuals and a strong R^2. For the group with follower counts between (50,000, 100,000] (Figure 13, Table 6), there was an insignificant change after the breakpoint (b_3), with a β_4 estimated at -13.251 and a p-value of 0.9010, so I failed to reject the null hypothesis for this group. For the group with follower counts between (100,000, 500,000] (Figure 15, Table 7), there was a significant β_4, estimated at -775.619 and a p-value of 9.8638E-08. The group with follower counts above 500,000 (Figure 17, Table 8) had the greatest change after the breakpoint (b_3), with a β_4 estimated at -1816.0393 and a p-value 1.8457E-08. For these 2 groups, I rejected the null hypothesis.

Figures 10, 11, & 12. Highest 3 Follower Groups Regression Residuals with One Breakpoint



Figures 13 & 14. Segmented Regression Model for Human Artists with (50,000, 100,000] Followers

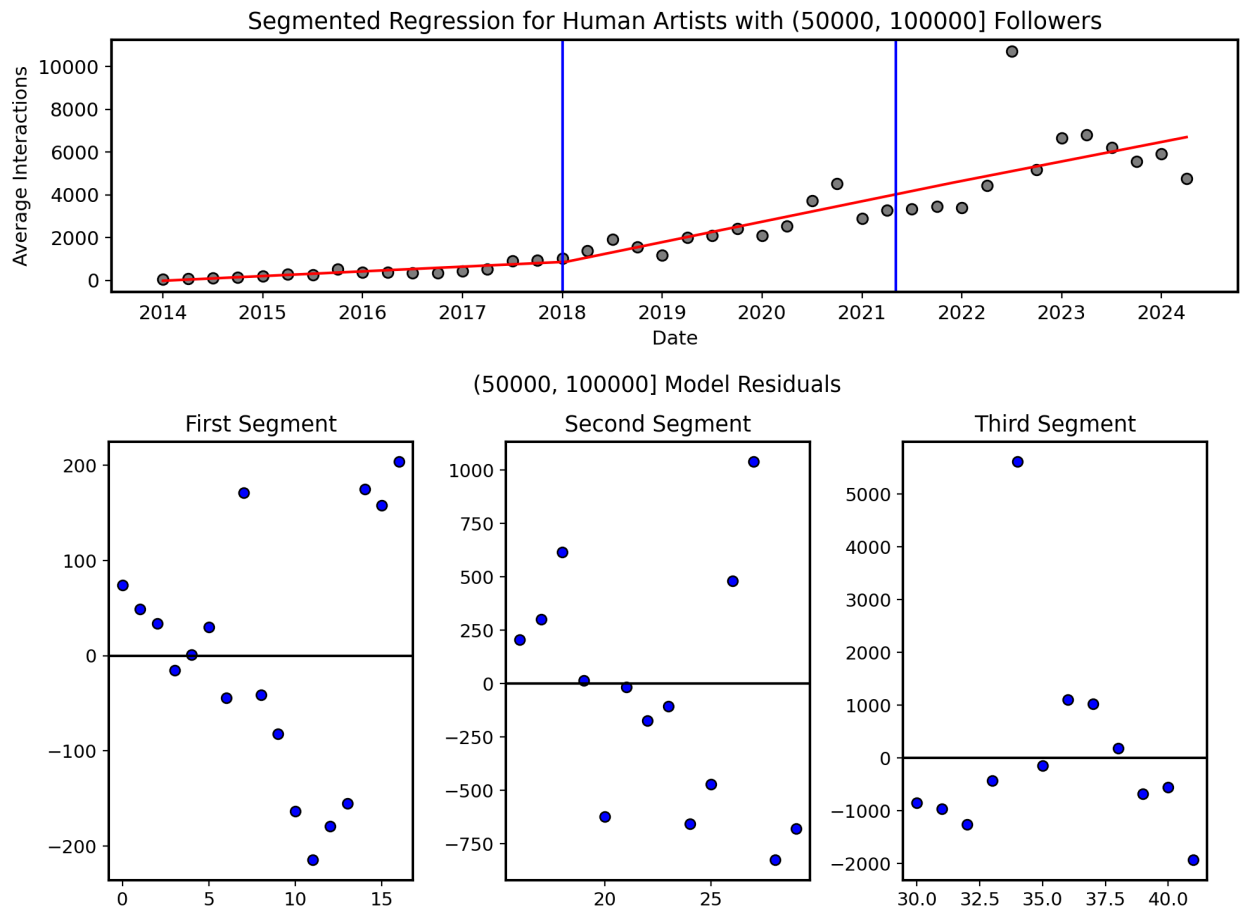


Table 6. Segmented Regression Results for Human Artists with (10,000, 50,000] Followers

No. Observations	42
No. Segments	3
SSR	46155652.52062782
SST	56908164.810386404
R^2	0.8110550159966479

Parameter	Estimate	Std. Err.	p-value
β_1	-19.657117757717334	489.34318408816733	0.9681675821796292
β_2	53.141641146504355	45.56523157492562	0.25077053184499787
β_3	187.89950805187325	84.45758654615733	0.03210999523293296
β_4	-13.250649781687097	105.80360804947428	0.9009952306044611
α_3	227.79049941669052	-	-
b_2	16*	-	-
b_3	29.35641897**	-	-

*Between 2017-11 to 2018-01, **Between 2021-04 to 2021-07

Figures 15 & 16. Segmented Regression Model for Human Artists with (100,000, 500,000] Followers

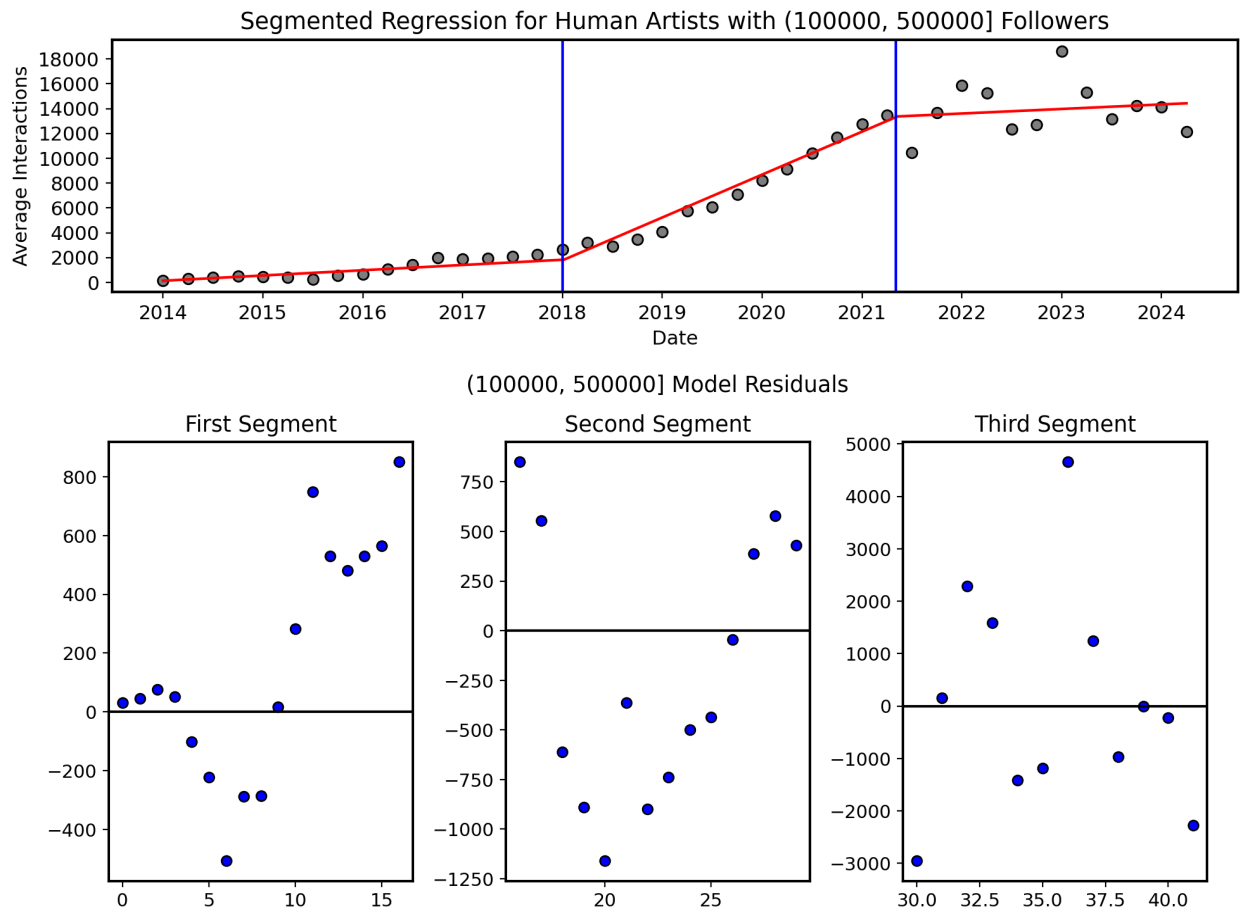


Table 7. Segmented Regression Results for Human Artists with (100,000, 500,000] Followers

No. Observations	42
No. Segments	3
SSR	57706938.963677995
SST	60181011.610254414
R^2	0.9588894805790394

Parameter	Estimate	Std. Err.	p-value
β_1	141.9814851953959	547.1609752559389	0.7966597445086074
β_2	102.5895434021358	50.94894004246919	0.05117572564275588
β_3	764.6385099587471	94.43657728362695	8.505562044836005E-10
β_4	-775.618885231478	118.30471384580854	9.863815793102342E-08
α_3	91.60916812940508	-	-
b_2	16*	-	-
b_3	29.35641897**	-	-

*Between 2017-11 to 2018-01, **Between 2021-04 to 2021-07

Figures 17 & 18. Segmented Regression Model for Human Artists with 500,000+ Followers

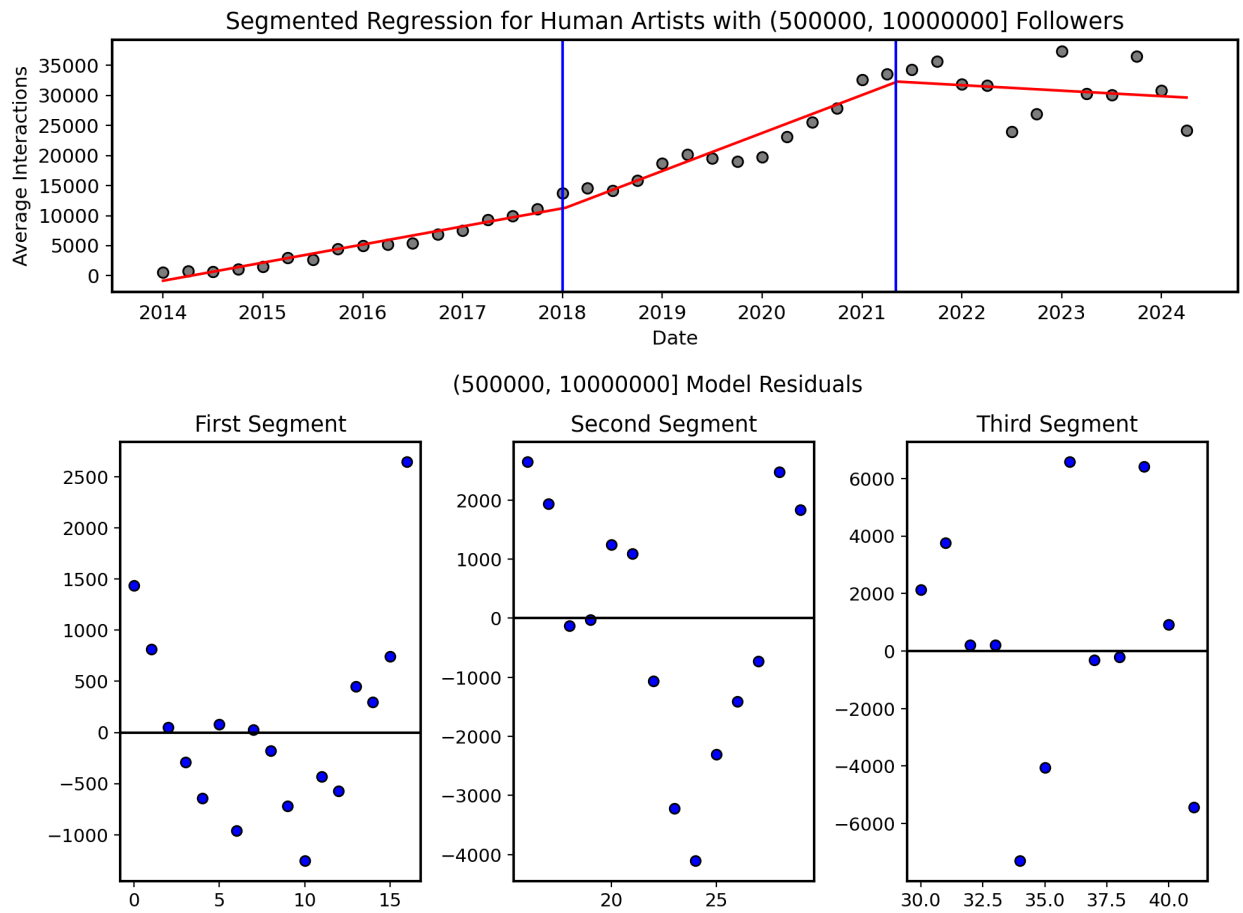


Table 8. Segmented Regression Results for Human Artists with 500,000+ Followers

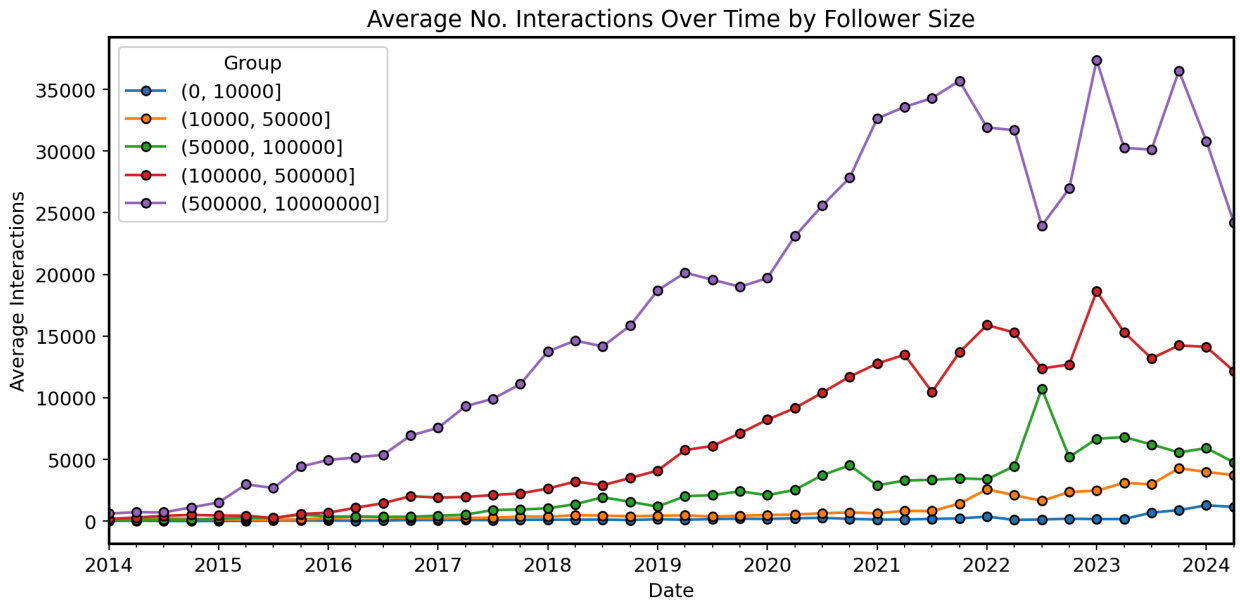
No. Observations	42
No. Segments	3
SSR	270338062.77315015
SST	283084392.8608264
R^2	0.9549733916488192

Parameter	Estimate	Std. Err.	p-value
β_1	-833.5754162440807	1184.280817039876	0.485807040797377
β_2	746.3480939100126	110.27440747686282	5.073520056181288E-08
β_3	840.9780504981173	204.3994947768147	0.00020082270497556536
β_4	-1816.0392766763878	256.0599339297675	1.845722837752586E-08
α_3	-228.71313226825774	-	-
b_2	16*	-	-
b_3	29.35641897**	-	-

*Between 2017-11 to 2018-01, **Between 2021-04 to 2021-07

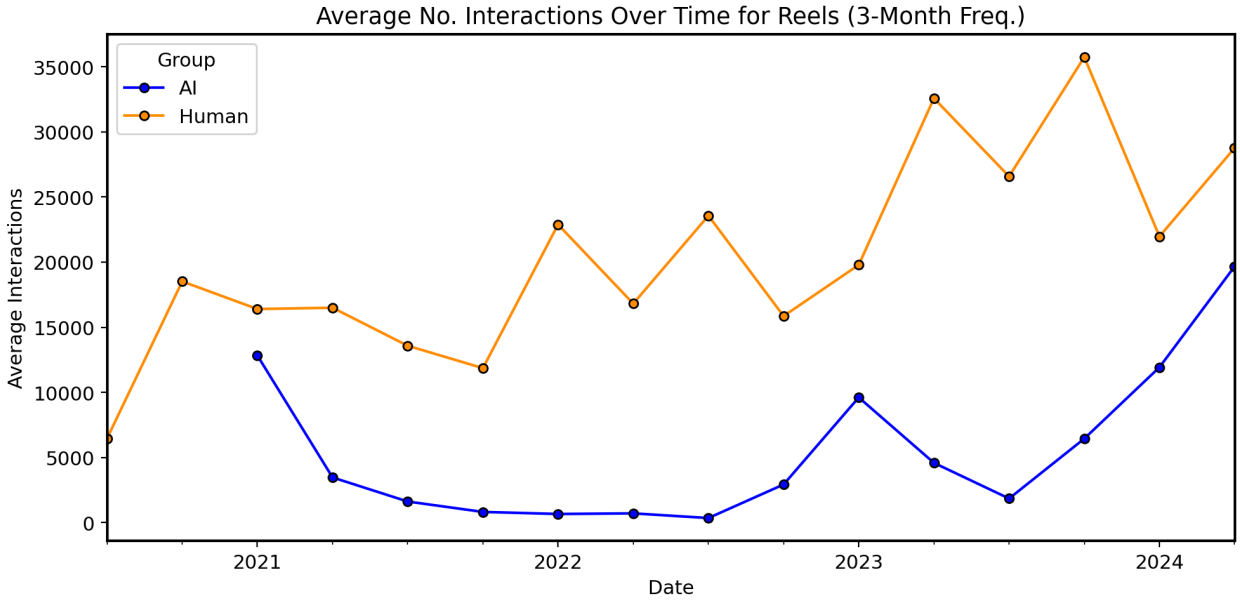
While the number of followers was only recorded at the time of data collection, the number of followers at one time point may be an appropriate estimator for the past number of followers. Based on Figure 19, the average interactions per month for the groupings based on the number of followers at one point are well separated. If the number of interactions is proportional to the number of followers, I can use this observation to claim that the follower groupings use appropriate ranked groupings for the accounts in the sample over time.

Figure 19. Average Interactions Over Time by Follower Size



Data was also collected on each profile's reel posts. These posts were categorized as reels at the time of sampling, and filtered out any posts before 2020, as Instagram only launched the reels feature in 2020. I initially explored this data, but no suspicion was apparent through visual inspection of the time series (Figure 20). For brevity, this analysis was excluded from this paper.

Figure 20. Average Interactions Over Time for Reels



DISCUSSION

The main aim of this paper was to investigate the recent change in trends of audience sizes for both AI art and human art posts on Instagram to determine a possible association between the two. This can be used to determine if human artists have experienced changes in recent years. Based on my findings, there was convincing evidence that both groups experienced significant change recently, with the decrease in human art growth rate occurring with the AI art increase. Human art growth decreased to the point that it started experiencing negative growth after July 2021. Given that AI art audience sizes started increasing almost at the same time, this proves my initial hypothesis that the growth rate of human art has had a negative association with the increase of AI art. When considering human artists individually, the artists most affected were the ones with high follower counts similar to those in my sample, specifically, those with higher than 100,000 followers, with indeterminable to little effect on artists with smaller follower sizes. While this may lead many to believe that AI art is the cause of the decline in human art

audiences, this paper does not provide enough evidence to show proof of a causal relationship. Despite this, the decline in audience size for human artists (similar to my sample) in recent years is not a result to be ignored. This unprecedented negative growth for human art as a whole may be a result of a devaluation of art in society, based on research by Hassine and Neeman (2019), and possibly caused less creative inspiration in other forms of art, such as in music and photography (Manovich, 2017). For the accounts with large followings, this has most likely affected their revenue streams, voice, and livelihood. Given the size of these affected artists, one can assume many of them make a significant portion of their income from their art profiles and other related means dependent on their audiences on Instagram, such as selling art courses or merchandise. A smaller audience also leads to a smaller voice for these affected artists. This leads to fewer sponsorships, less activism, and a smaller community surrounding and benefiting these human artists and their followers, according to Li & Kang (2019). These findings contribute to filling the targeted gap by establishing foundational trends in audience sizes of human and AI art, as well as providing methods to discover them. Keeping these results in mind, there could be many explanations for this association in trends, but determining these is outside the scope of this paper.

Limitations:

Since the profiles were selected from Instagram and Google searches, it is likely that bias was introduced into the sample. This convenience sample possibly biased profiles to be ones that appeared more in searches, and had attributes that search engines value, such as popularity, recency, owner's post schedule, time of day, location, how well posts use keywords, and other factors that the Instagram search algorithm uses, as well as Google. This may have made my data include profiles that have posts with high interactions, engagement rates, or recent growth. While

there was an attempt to minimize this bias with the diversification of follower sizes, it was not a substitute for randomness.

The Instagram algorithm reflects many underlying processes online, leaving causal relationships indeterminable. While this could have been overcome with a control for human art unaffected by AI to attribute changes solely to AI art, this is not feasible because this is a historical effect that's difficult to simulate given the connected and hidden nature of Instagram. Some factors cannot be accounted for as the Instagram algorithm has not been studied frequently or thoroughly, and its criteria are not publicly available.

This study would benefit from other performance metrics, but that information is not publicly available to people who are not the owners of the posts. This data would include the number of shares, development of each post metric over time (instead of one snapshot of it), number of accounts reached, number of accounts engaged, and demographics of the audience. This would allow my definition of the audience to be much more accurate, as well as give me better data to possibly employ more advanced quasi-experimental methods, such as with deep learning networks. The limited data I had and my limited ability led me to settle for the segmented regression model.

Finally, I did not check if all posts from the profiles in the sample were representative of the groups they were in. Instagram profile feeds are sorted in reverse chronological order, leaving old posts at the bottom. While I manually checked if most of the recent posts in profiles were similar to the art genre I classified them as, less attention was given to older posts, especially if artists posted frequently and their feed spanned dozens of pages, as it would take an exorbitant amount of time for all 256 profiles and dozens of additional candidates for the sample analyzed by one person. One result of this is possible erroneous categorizations of AI art further

back in time. I categorized some old posts as AI since most of the profile's recent posts were, but given that some of these posts are from multiple years before AI art was available to the public, these most likely were not AI art posts and were not representative of its audience size. There was an attempt to minimize this by keeping the scope of the analysis after 2014, but this does not account for multiple years between then and the boom of AI art, without cutting out valuable human art data.

Future Research:

The topic of AI and Human art online is very broad and has many possible avenues for further research. I suggest researchers investigate the aesthetic quality of images generated by both groups, monetary gains gained by artist profiles, tendencies of the Instagram algorithm, and to use more data with the cooperation of artists. Using more data will allow researchers to employ advanced quasi-experimental methods to establish causality. The results from the analysis on Instagram provide reason to believe that these trends have occurred on similar social media platforms. I also suggest conducting similar studies on other social media platforms with significant art uploads, such as Twitter and Pinterest. This will start addressing the possible differences in online platforms and implications for online artists in general.

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APPENDICES

Appendix A: Table of All AI Artist Usernames in Sample

Genre	1,000 - 10,000	10,000 - 50,000	50,000 - 100,000	100,000 - 500,000
Anime	fuurin_aiart	the.artful.ai	sushi_musume_jk	brunirax.digitalart
Anime	wellingtonton29	asmi_creator	anime00s	bebop.genesis
Character Design	midjourney_mondays	raymur.art	roguerobot.ai	aicharactersart
Character Design	ai_hiro_abe	kaptaincoca	bra1nst0rming_	null
Environment	prompt.soru	mystics_meta	wh3re.is.the.sun	eightbitstrana
Environment	ai_equiped	iguana_nick	iamsentient_	darylanselmo
Fantasy	mdjrn.charmings	alvia.art	ephellem	twistedvisions.ai
Fantasy	dreamerszenltd	majara__	dreamworlddweller	digital_art_fusion
Fashion	modelaindo	glamoph	doopiidoo	tinkertailorart
Fashion	mekuwelt	fashion.coupids	bonnycarrera	aidesign.png
Horror	hellimagery	thehorror.ai	art_from_the_aether	chrisperna
Horror	goat_cigs	macabre.masters.ai	neural.horror	null
Mecha	clueless.machines	igor_art_ua	hxai_digitalart	null
Mecha	ankan_bhowmick.ai	kzy2017	null	null
Portrait	arttacksyou	mako_aiart	imaginationtwist.ai	mindful.machine.art
Portrait	ekjahanor2021	girolamoartimagery	niyfulx	artworks_by_rokii
Realistic Anime	nao_yukiyanagi	akenoslove	bluekitsune.ai	anime.loveii
Realistic Anime	bandc.ai	aidolsmedia	freiart_mjr	freiart_mjr
Semi Realism	impish.ai	pixiewonderland.ai	daydream.ai	nephtys__s
Semi Realism	imvidvn	joerussketchbook	_artsopolis	rowan_lewgalon_aiart
Stylized	midjoeney	tonybamber	24kenva	diegosilvadigital
Stylized	illusionoflaife	zenithworkshop	jeaninespets	3lot3ro
Traditional Painting	ylmz_omer32	biohacker	cy_artsfun	01000101_01100111_01101111
Traditional Painting	vince.hogge	marilyns_canvas	sunflowers_ai	lostdreams.ai

Appendix B: Table of All Human Artist Usernames in Sample

Genre	1,000 - 10,000	10,000 - 50,000	50,000 - 100,000	100,000 - 500,000	500,000+
Anime	yukari_ow	lewillust	ichigowarano	yoneyamai	null
Anime	reia_76	yugan.999	tamaji599	deariyue	kuvshinov_ilya
Character Design	seth_pennington_	polluxery	cherryandsisters	hou_jae04	koyori_n
Character Design	birmit_c	shlerbe.art	julbenart	onebadnoodle_	null
Environment	changwei_chen_art	jtuffdesigns	dewpearl_art	cordeliacarla	paulkenton
Environment	alexshin_art	djosephcreative	milicamastelica	juliaspowellart	markmaggiori
Fantasy	jingliang_j	gabrielleragusi	theartofmichaelwhelan	tahraart	anniestegg
Fantasy	barret_frymire	phil__moss	nn__ee__zz	ryanhappytree	beksinski
Fashion	jacquelineostermann	myrtostamou	nam.felixchen	eris_tran	hnicholsillustration
Fashion	junztv	kuzminanastya	tess.laf	ahvero	null
Horror	ginettarah.art	mr.deadwater.art	gravemud.art	stitching_a_laugh_to_darkness	laurasbeatingart
Horror	m_g_t_art	builtfromsketch	anna.scary.art	justinonealart	null
Mecha	moyan_0802	tg_ping	fightpunch	toyagogo	pascalblanche
Mecha	dickdavid_sketches	estevesdesign	freakyfir	emersontung	null
Portrait	kieracoyleart	drafts_jotape	mahiartss	9brookskim	stephenbaumanartwork
Portrait	anjelo_art	sevdehrart	bochan_0619	mysketchvetch	devonrodriguezart
Semi Realism	sybillgray	phamoz	svanhaart	hessah._.art	lotusbubble
Semi Realism	_lucidream_	aikee.png	_siimka._	chris_firger	samdoesarts
Stylized	drawligator	elfphoe	_spec	o.sporin	carolynj
Stylized	fizzeru	crystalsugarstars	crazycomet107	sykosan	gretlusky
Stylized	freethespice	kanioque	miithers	friedbaens	alaiganuza

Traditional Painting	danielfolta	rakhmanovsamir	tashkotwal	rossoartist	andrewcadima
Traditional Painting	shelleyhannaart	carolinenelson.art	estelledayfineart	brian_neher	mcfriendy
Cartoon	elinaclevergull	goodbyellow	pamekute	idolomantises	feefal
Cartoon	steamshade_art	cardbordtoaster	hallojigsawpanda	varguyart	blobbynfriends
Comic	redrollerball	jhonycaballeroart	dd.markk	massiveface	thealexrossart
Comic	calebandersonart	luisbajocollados	katekomics	real_greg_capullo	jimlee
Digital Painting	kamyarlabbaf	mahonsunto	levinky_art	_alariko	wlop
Digital Painting	afique_asyraf	rxzarx	percichen	jocelincarmes	yuumeiart
Pencil	__shashi__artist__4436	art._.verse	jahzeelv_jw	maas.art	chommang
Pencil	monotanix	sachinbhojane_art	underwaterlad	sinarty77	eleeza
Pixel	angrymonsterham	transientcodepixelart	nop_pixels	8pxl_	pixel.bef
Pixel	hby_stuff	pixelianska	angrysnail	woostarspixels	null